

# Topology-Conditioned Interference and Resource-Isolation Trade-offs in 5G NR Sidelink UAV Swarms

Panagiota Grosdoulis

Department of Electrical and Computer Engineering  
Democritus University of Thrace  
Xanthi, Greece

**Abstract**—Direct NR sidelink is a candidate for infrastructure-independent UAV-swarm communication, but dense aerial line-of-sight conditions make scalability depend on more than swarm size alone. We present a measurement-grounded, NR-aware system-level study of communication topology, exact frequency-resource partitioning, and directional desired/interferer selectivity. A 100-seed campaign evaluates  $N \in \{5, 10, 20, 30, 50, 75, 100\}$ , partitions a 133-PRB profile over  $R \in \{1, 2, 4, 8\}$  abstract orthogonal resources, and tests  $G \in \{0, 3, 6, 9\}$  dB relative desired/interferer advantage. Under the full-load sequential-disjoint baseline,  $N = 100$  yields  $-23.03$  dB mean SINR, 0.0128 first-transmission success probability, and 0.359 Mbps expected PHY goodput. Yet greedy short-link pairing at the same  $N, R = 1, G = 0$  reduces mean desired-link length from 517 m to 84 m and raises success probability to 0.4766 and goodput to 15.116 Mbps. Resource partitioning and spatial selectivity expand the evaluated reliability-goodput region, but partitioning exhibits a non-monotonic interference/bandwidth trade-off. The results therefore support topology-conditioned operating regions rather than a universal density-only swarm-capacity claim.

**Index Terms**—5G NR sidelink, UAV swarm, topology, interference, resource isolation, operating region, reproducibility.

## I. INTRODUCTION

Reliable wireless links are a core enabling technology for cooperative UAV systems, whose mobility and altitude create propagation, interference, and networking conditions that differ substantially from terrestrial deployments [1], [2]. NR sidelink provides standardized direct-communication mechanisms [3] and is therefore a relevant candidate for intra-swarm communication, as already explored in UAV-swarm scheduling and suitability studies [4], [5].

Aerial line-of-sight can strengthen both the desired link and many simultaneous interferers. Consequently, increasing  $N$  can push a shared resource toward aggregate-interference domination, but  $N$  alone is not sufficient to predict the regime: the desired-link geometry induced by the communication topology can materially change the SINR distribution. Interference mitigation is also cross-layer. Frequency-resource isolation reduces co-channel interference but simultaneously reduces per-link PRBs and transport-block size, while spatial selectivity changes the desired/interferer power ratio without adding bandwidth.

Prior work covers sidelink scheduling, analytical U2U outage, joint trajectory/frequency/routing optimization, topology

control, MCS adaptation, and synchronization [6]–[10]. What remains insufficiently characterized is their intersection: a controlled NR-aware operating-region study that conditions density effects on communication topology while explicitly charging resource isolation for its PRB/TBS cost.

This paper asks: *How do communication topology and swarm density jointly determine the evaluated UAV-sidelink interference regime, and when do bandwidth-aware resource isolation and spatial desired/interferer advantage expand the resulting reliability/goodput operating region?* The contributions are threefold. First, we combine measurement-derived A2A path loss with 3GPP-derived MCS/TBS/LDPC mechanics under an explicit study resource grid and sourced 5G-LENA link-level BLER curves. Second, matched-seed sequential versus greedy short-link pairing establishes topology as a first-order scaling variable in the evaluated model. Third, we quantify the resource-isolation trade-off and joint operating envelope while varying allocator, link adaptation, and noise-bandwidth accounting. The paper is intentionally system-level and does not claim a standards-complete sidelink PHY/MAC or measured multi-UAV RF performance.

## II. RELATED WORK AND RESEARCH GAP

UAV-communications surveys identify line-of-sight propagation, three-dimensional mobility, interference, and topology as central design variables [1], [2]; the NR-V2X literature documents sidelink physical-layer and autonomous resource-selection mechanisms [3]. UAV-specific work uses sidelink for communication-aware scheduling and multi-hop forwarding [4], evaluates NR sidelink for internal swarm communication [5], and studies dynamic U2U MCS control in ns-3 [8]. Complementary work derives multi-UAV interference/outage models [6], jointly optimizes trajectory, frequency, and routing [7], or targets synchronization and Doppler-aware sidelink access [10].

Topology itself is also an active optimization variable. Bai et al. jointly optimize UAV connection relationships, transmit power, traffic, and positions under mutual interference [9]. We therefore do not claim that topology control is new. The narrower gap addressed here is an NR-aware, matched-seed characterization of how desired-link topology changes the density/interference regime and how exact resource isolation

behaves once its reduced PRB/TBS budget is propagated through link adaptation and BLER mapping.

Our study accordingly does not propose a standards-complete sidelink protocol. It isolates swarm size, desired-link topology, frequency-resource partitioning, a geometry-based reuse allocator, and an experimental spatial desired/interferer advantage. Pairing, random versus geometry-aware allocation, adaptive versus fixed MCS, and noise-bandwidth accounting are varied explicitly so that model-sensitive conclusions are exposed rather than hidden inside a single optimized policy.

### III. SYSTEM MODEL AND REPRODUCIBILITY

#### A. Propagation and link abstraction

The large-scale A2A channel uses the 3.5 GHz measurement-derived fit of Erdemir *et al.* [11],

$$PL(d) = 34.650 + 21.66 \log_{10}(d/1 \text{ m}). \quad (1)$$

Simulated RF values are model-derived, not field measurements. The study profile uses 50 MHz, 30 kHz SCS, 133 PRBs, and 0.5 ms slots. The 133-PRB bandwidth configuration is consistent with the NR UE table [12], while the 0.5-ms slot duration follows the 30-kHz NR numerology [13]. MCS values, TBS quantization, and LDPC segmentation follow NR procedures [14], [15], but the resource grid is an explicit study profile rather than a complete PSSCH implementation: each slot reserves two control/PSCCH-like symbols and one guard symbol, uses 11 PSSCH symbols, and removes 24 DM-RS RE/PRB. PSFCH, SL-PRS, and detailed first/second-stage SCI RE accounting are not modeled explicitly.

For each candidate MCS, the computed TBS determines base graph and nominal code-block size. The requested code-block size is mapped to the nearest available 5G-LENA curve with the same MCS/base graph; BLER is linearly interpolated in SINR and clamped to the sourced curve endpoints. Transport-block BLER is derived from code-block BLER under an independent-code-block approximation. These are generic NR EESM link-level simulation tables from 5G-LENA [16], [17], not measured UAV or PSSCH-specific BLER data. Link adaptation selects the available MCS maximizing expected first-transmission delivered bits per slot; a fixed-MCS-4 control tests this optimistic instantaneous-SINR assumption.

#### B. Topology and interference controls

For swarm size  $N$ , the simulator forms  $\lfloor N/2 \rfloor$  one-way disjoint transmitter–receiver links; all formed links transmit simultaneously in each static snapshot and one UAV is unpaired when  $N$  is odd. The baseline pairs node indices  $(0, 1), (2, 3), \dots$ , so desired-link distance is independent of proximity. The robustness topology instead globally sorts pairwise distances and greedily selects the shortest non-overlapping pairs; we call this *greedy short-link pairing*, not a standardized association rule.

The 133 PRBs are partitioned as evenly as possible across  $R \in \{1, 2, 4, 8\}$  abstract frequency resources. Different resources are assumed perfectly orthogonal, i.e., no adjacent-channel/inter-carrier leakage is modeled. For  $R > 1$ , the

TABLE I  
BASELINE STUDY CONFIGURATION.

| Parameter                 | Value                                           |
|---------------------------|-------------------------------------------------|
| Area / altitude           | 1000 × 1000 m / 100 m                           |
| Carrier / bandwidth / SCS | 3.5 GHz / 50 MHz / 30 kHz                       |
| Tx power / noise figure   | 30 dBm / 7 dB                                   |
| Grid profile              | 133 PRBs; 2 control + 1 guard; 11 PSSCH symbols |
| Active links per snapshot | $\lfloor N/2 \rfloor$ , one-way, full load      |
| Swarm sizes $N$           | 5, 10, 20, 30, 50, 75, 100                      |
| $R / G$                   | 1, 2, 4, 8 / 0, 3, 6, 9 dB                      |
| Monte Carlo               | 100 matched deterministic seeds                 |

baseline reuse allocator forms the symmetric geometric conflict weight

$$w_{ij} = d^{-2}(\text{Tx}_i, \text{Rx}_j) + d^{-2}(\text{Tx}_j, \text{Rx}_i), \quad (2)$$

orders links by total conflict weight, and greedily assigns each link to the resource with minimum accumulated same-resource weight. This is a research abstraction, not NR Mode-1/Mode-2 sensing or scheduling [3]. The inverse-square conflict proxy is intentionally heuristic and is not identical to the fitted A2A path-loss exponent 2.166 used by the propagation model. Seeded random assignment is the no-coordination control. Directional relative advantage  $G \in \{0, 3, 6, 9\}$  dB is imposed as  $+G/2$  dB desired gain and  $-G/2$  dB co-channel interference gain; it is a sensitivity variable, not a beamforming model.

#### C. Metrics, statistics, and evidence

First-TX success probability is  $1 - \text{TB- BLER}$ . Expected PHY goodput is  $\text{TBS}(1 - \text{TB- BLER})/0.5$  ms under one scheduled TB per active link per slot; it excludes queues, higher-layer overhead, retransmission timing, and scheduler idling. Primary treatment comparisons reuse identical seeds/geometries and report paired differences, Cohen’s  $d_z$ , and paired percentile-bootstrap 95% intervals computed from 10,000 resamples with a fixed RNG seed for reproducibility.

The operating envelope is an explicit engineering-policy summary: a scenario is *mean-policy-qualified* when its scenario-mean first-TX success probability is at least 0.10 and mean expected PHY goodput is at least 1 Mbps. Thresholding uses mean point estimates, not lower confidence bounds, so the reported largest evaluated  $N$  is not a capacity guarantee.

Code, deterministic seed definitions, compact publication extracts, and workflow provenance are versioned in the public repository,<sup>1</sup> while full per-seed evidence is archived by the GitHub Actions artifact recorded in the repository. The audited publication run used commit `f224ba9e5f95` and stores the manuscript snapshot, generated tables, and artifact digest together.

### IV. RESULTS

#### A. Baseline interference and resource partitioning

Under sequential-disjoint pairing with one shared resource and  $G = 0$ , mean SINR decreases from  $-0.49$  dB at  $N = 5$  to  $-23.03$  dB at  $N = 100$ . At  $N = 100$  (50 simultaneous directed

<sup>1</sup><https://github.com/panagiotagrosdouli/uav-sidelink-swarm>

TABLE II  
SELECTED  $N = 100$  BASELINE OPERATING POINTS.

| $(R, G)$ | SINR (dB) | Success prob. | Goodput (Mbps) |
|----------|-----------|---------------|----------------|
| (1, 0)   | -23.03    | 0.0128        | 0.359          |
| (2, 0)   | -16.20    | 0.0749        | 0.668          |
| (4, 0)   | -10.48    | 0.0775        | 0.572          |
| (8, 0)   | -5.47     | 0.2049        | 1.042          |
| (8, 6)   | 0.53      | 0.5292        | 2.229          |
| (8, 9)   | 3.53      | 0.8050        | 3.493          |

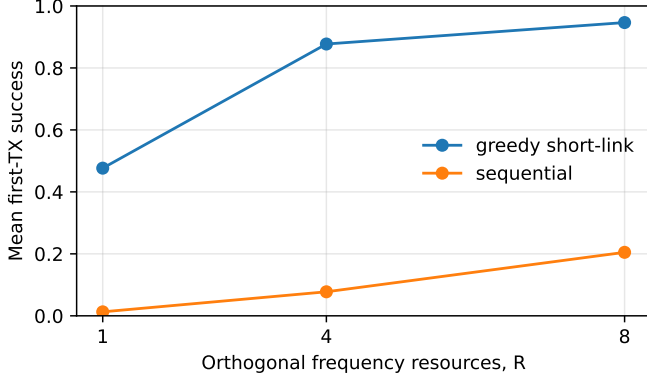


Fig. 1. Pairing sensitivity at  $N = 100, G = 0$  with adaptive MCS and conflict-graph allocation.

links), first-TX success probability is 0.0128 and expected PHY goodput is 0.359 Mbps. Using the explicit success-probability threshold 0.5, the mean per-seed conditional share among policy-failed links is 72.4% aggregate-interference limited and 27.6% dominant-interferer limited. This is a heuristic diagnostic, not a pooled-link fraction or a causal channel classification.

Table II shows that frequency-resource separation can improve reliability despite reducing the PRBs available to each link. Goodput is non-monotonic:  $R = 4$  has higher mean SINR than  $R = 2$  but lower goodput. This reflects the joint effect of lower co-channel interference, changed conflict-graph assignments, and smaller per-link TBS; more resource groups do not guarantee more goodput. With  $R = 8, G = 6$ , first-TX success probability reaches 0.5292 and expected goodput 2.229 Mbps. Relative to  $R = 1, G = 0$  at  $N = 100$ , the matched-seed differences are 0.5164 in success probability (paired-bootstrap 95% CI [0.5060, 0.5266],  $d_z = 9.61$ ) and 1.870 Mbps in goodput (CI [1.721, 2.006],  $d_z = 2.59$ ).

#### B. Topology is a first-order determinant in the model

At  $N = 100, R = 1, G = 0$ , sequential pairing produces a 517.20 m mean desired-link distance, whereas greedy short-link pairing reduces it to 84.05 m. Mean SINR changes from -23.03 to -1.75 dB, first-TX success probability from 0.0128 to 0.4766, and goodput from 0.359 to 15.116 Mbps. Fig. 1 shows that the difference persists as  $R$  increases. Thus, the severe degradation of the sequential baseline is not a topology-independent density law.

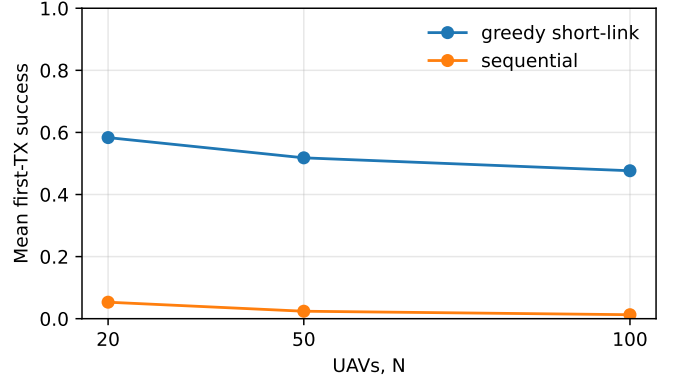


Fig. 2. Topology-conditioned scaling for  $R = 1, G = 0$ .

TABLE III  
ROBUSTNESS CHECKS AT  $N = 100$ .

| Dimension                       | Control / treatment                   | Main observation                       |
|---------------------------------|---------------------------------------|----------------------------------------|
| Pairing, $R = 1, G = 0$         | sequential $\rightarrow$ short-link   | success 0.0128 $\rightarrow$ 0.4766    |
| Allocator, $R = 8, G = 0$       | random $\rightarrow$ conflict graph   | goodput 0.815 $\rightarrow$ 1.042 Mbps |
| Link adaptation, $R = 8, G = 6$ | fixed MCS-4 $\rightarrow$ adaptive    | goodput 0.722 $\rightarrow$ 2.229 Mbps |
| Noise bandwidth                 | nominal share $\rightarrow$ exact PRB | max $ \Delta \text{SINR}  = 0.113$ dB  |

The same effect appears across swarm size. With  $R = 1, G = 0$ , sequential pairing keeps mean desired-link distance near 515–525 m from  $N = 20$  to 100, while success probability falls from 0.0531 to 0.0128. Greedy short-link pairing reduces mean desired-link distance from 195.3 m at  $N = 20$  to 84.0 m at  $N = 100$ ; success probability changes from 0.5834 to 0.4766 despite the increasing number of simultaneous links (Fig. 2). The experiment therefore separates two competing geometric effects: more active transmitters increase interference, while denser local pairing strengthens the desired link.

#### C. Allocator and link-adaptation controls

Resource coordination adds benefit beyond increasing  $R$ . At  $N = 100, R = 8, G = 0$ , sequential pairing gives success/goodput 0.2049/1.042 Mbps with the conflict graph versus 0.1463/0.815 Mbps with random allocation. Under greedy short-link pairing, the corresponding values are 0.9465/10.836 and 0.8824/8.357 Mbps. Link adaptation is also material: at  $N = 100, R = 8, G = 6$  under sequential pairing, adaptive MCS gives 0.5292/2.229 Mbps versus 0.3309/0.722 Mbps for fixed MCS-4. The adaptive result is therefore an optimistic system-level control case, not normative 3GPP AMC. By contrast, changing from nominal-channel PRB-share noise bandwidth to exact occupied PRB bandwidth changes SINR by at most 0.113 dB over the overlapping robustness grid, with negligible success/goodput change.

#### D. Mean-based policy operating envelope

For the explicit policy requiring scenario-mean first-TX success probability  $\geq 0.10$  and mean expected goodput

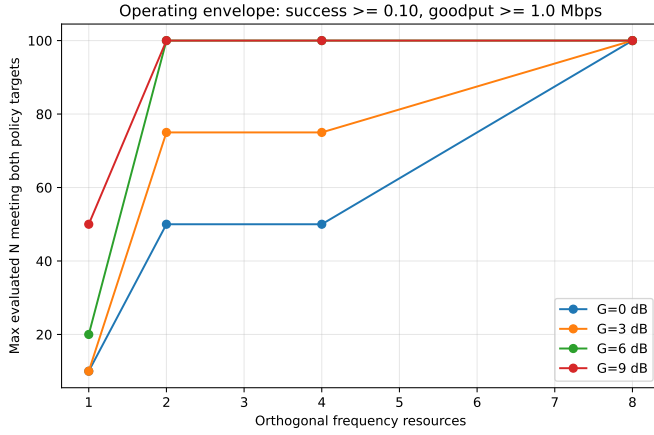


Fig. 3. Mean-based operating envelope for the sequential baseline under explicit policy thresholds.

$\geq 1$  Mbps, Fig. 3 reports the largest *evaluated*  $N$  meeting both point-estimate thresholds for each  $(R, G)$ . The sequential baseline reaches  $N = 10$ ; several resource/directionality combinations reach the evaluated  $N = 100$  boundary. These points are policy-qualified operating points, not confidence-bound guarantees or universal capacity limits.

## V. DISCUSSION AND LIMITATIONS

The results show that topology and resource reuse cannot be separated when interpreting dense aerial sidelink. Prior multi-UAV work likewise couples interference with geometry, routing, and frequency allocation [6], [7]; recent U2U sidelink work also demonstrates that MCS adaptation materially affects throughput/reliability trade-offs [8]. Our controls show the same dependence: resource partitioning helps the long-link baseline, but shortening desired links changes the shared-resource regime more strongly.

Orthogonalization is not free. The  $R = 2$  versus  $R = 4$  result reflects the joint effect of conflict-graph placement, lower co-channel interference, and reduced TBS as PRBs are split. Accordingly, the operating envelope is a map for one explicit topology/load/model combination rather than a scalar maximum swarm size.

### A. Threats to validity

*Propagation domain:* the fitted A2A law comes from a 3.5 GHz, 50 MHz, 30 dBm campaign with both drones at 100 m, an initial separation of 85 m, and a roughly 1 km A2A flight path in a mostly hilly/forested, rural-like campus environment [11]. Our carrier, bandwidth, altitude, and transmit power match those conditions, but the synthetic square permits link lengths up to 1.41 km, whereas the measurement campaign reports a 1-km transmitter trajectory starting from 85 m separation. The longest simulated links may therefore extend beyond the measurement campaign’s sampled separation geometry. Shadowing correlation, fast fading, blockage, Doppler, antenna-body effects, and temporal channel memory are omitted.

*PHY/MAC abstraction:* the study uses 3GPP-derived MCS/TBS/LDPC rules but an explicit simplified resource grid. PSFCH, SL-PRS, detailed SCI resource accounting, feedback timing, retransmissions, power control, and standards-complete sensing/resource selection are outside the paper model. Different abstract frequency resources are perfectly orthogonal. The 5G-LENA curves are generic NR EESM link-level tables [16], [17]; requested code-block size is mapped to the nearest sourced same-MCS/base-graph curve, SINR is linearly interpolated and endpoint-clamped, and TB-BLER uses an independent-code-block approximation. These choices bound the absolute goodput interpretation.

*Traffic/topology abstraction:* each snapshot contains  $\lfloor N/2 \rfloor$  simultaneously active one-way links; it does not model bidirectional application traffic, queues, mobility during a packet exchange, or mission-constrained partner selection. Greedy short-link pairing is a sensitivity control, not a proposed association protocol. Likewise,  $G$  is a desired/interferer power-ratio sensitivity, not an antenna-array or beam-tracking model.

*Statistics and policy:* matched seeds reduce geometry variance in treatment comparisons, but the operating-envelope thresholds are engineering choices applied to scenario means. They do not require the lower confidence bound to satisfy the target and therefore should not be interpreted as reliability guarantees.

### B. Design implications

Within these boundaries, four conclusions are supported by the evaluated model: (i) desired-link topology must be specified before interpreting density scaling; (ii) more frequency groups do not necessarily improve goodput because interference reduction competes with PRB/TBS loss; (iii) geometry-aware reuse outperforms random allocation in the tested cases; and (iv) link adaptation and spatial selectivity complement rather than replace topology/resource design. Future work should repeat the matched-seed methodology with standards-complete sidelink sensing/scheduling, richer channel dynamics, realistic bidirectional traffic, and multi-UAV RF validation.

## VI. CONCLUSION

Under the evaluated full-load sequential-disjoint topology, increasing  $N$  increases the number of simultaneous directed links and drives the shared-resource case into a severe interference-limited regime. Resource partitioning and the experimental desired/interferer advantage move the mean-based operating region substantially. The robustness campaign also shows why this result must be topology-conditioned: at  $N = 100$ , greedy short-link pairing raises shared-resource first-TX success probability from 0.0128 to 0.4766 by shortening desired links. Geometry-aware reuse and adaptive MCS provide additional gains, while the tested noise-bandwidth convention has negligible effect. The supported design message is therefore conditional but actionable: topology, reuse, spatial selectivity, and link adaptation should be evaluated jointly rather than inferring swarm capacity from  $N$  alone.

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